The battle with Anopheles Gambie in Brazil: pure luck or successful interventions.

Image 1: map of South America showing areas in Brazil that were affected by Anopheles Gambie was established and eradicated in 1930. Map A shows suitable areas of Anopheles arabiensis and map B shows suitable areas of Anopheles gambiae in South America. The square seen northeastern of Brazil on both maps shows the area where members of Anopheles gambiae complex were found and eradicated around 1940.

Introduction

Anopheles gambie is a type of anthropoid that falls under the family Culicidae, their morphological characteristics include sensory organs near the mouth parts called palps that are similar in length as its proboscis (elongated sucking mouthpart), and their wings are covered with discrete blocks of black and white scales (Dujardin, 2017). They are commonly known as a ″vector″ of transmitting malaria because of their ability to carry a disease from one human to the other. Humans and female Anopheles mosquitoes are cyclically infected during malaria's natural history. In humans, - the parasites first develop and multiply in the liver cells and later in the red blood cells. Consecutive parasitic broods develop inside red blood cells in the blood and kill the cells, producing daughter parasites ("merozoites") that invade other red blood cells to continue the cycle. Malaria symptoms are brought on by the blood stage parasite, when the mosquito feeds on an infected person, the parasite is transmitted to the mosquito and it takes 10 to 18 days to migrate to the salivary glands of the mosquito and the infection cycle continues (A.James, 2020).

A, gambiae was initially identified in Brazil in March 1930 by entomologist Raymond C. Shannon of the Rockefeller Foundation and a member of the Cooperative Yellow Fever Service in Natal, when over 2000 larvae were found within a kilometre of the port of Natal, where mail ships from west Africa frequently anchored. The SCFA and the governor of Rio Grande do Norte at the time were alerted by the first epidemics in Natal in early 1930. Through a quick, palliative response by this agency, the mosquito was eradicated from the Natal urban region in 1932, but no more actions followed this first effort. Six years before A, gambiae was driven out of Natal, it spread throughout the interior regions of the states of Rio and Ceará leading to the biggest malaria epidemic in the Americas. This mosquito spread ″silently ″in the early 1930, took unpredicted importance to public health priorities 9 years after it was discovered. It was referred to as "the worst immigrant that could reach us from the north" by Afrânio Peixoto in 1940 (Gabriel, 2018)

The rapid organization of a new antimalaria agency was ordered by Brazil's President on August 5,

1938, to address this situation. Although spraying of homes, cars, planes, and boats was done to kill adult mosquitoes, most of the efforts was focused at eradicating their larvae. The Malaria Service of the Northeast started its operations in 1939 with no very specific plan for the eradication of gambiae from Brazil, aside from relying on Paris Green, and starting from the edges of the infested area toward the centre. Paris Green could be sprayed quickly when combined with dust or sand and was highly successful at eliminating larval anopheles. Large number of field workers were each allocated potential breeding sites to supervise. The Malaria Service of the Northeast's efforts were meticulously planned and well defined, which was the fundamental key to its success. An aerial photography-based cartography unit was quickly established, and the afflicted region was thoroughly surveyed and remapped. adulticide and medicinal measures were put in place to help those in need. Surprisingly, the effort to eradicate malaria from Brazil was well under way by the middle of the 1940s, and the disease has been successfully suppressed in Brazil for more than 20 years (G.F. Killeen, 2002).

The question now comes as whether the local climate in Brazil helped the government's 1930s efforts to contain and eradicate Anopheles gambiae. In this report I aim to show and express my opinion on how climate aided Brazil in the eradication of the species.

Methods
In the practical the following were provided:
Climax for windows computer program
This is a computerised system that analysis the likelihood that a species of plant or animal would survive in a specific area due to the climatic conditions of that area. We used map climex to generate data such as EI (Ecoclimatic Index), annual rainfall, the annual relative humidity at 9 am and 3 pm and the annual temperature of regions in Brazil and Africa. By using this data, we were able to compare the two location, along with some other regions in Brazil such as Natal, Dakar and Monte Santo to analyse the likelihood of Anopheles Gambiae surviving in this regions.

Article by Killen et al named 2002 Anopheles gambiae in Brazil, “The rise and fall of A gambiae in Brazil” as well as other papers related to the topic.
The articles focuses on the the discovery, spread and successful eradication of the A. Gambiae from Brazil. They mostly emphasis the measures that the brazil authorities took to ensure that the species was eradicated in region that were most affected such as Natal and Dakar.

Pros

- Can predict climate conditions over longer time scales. By using this model, we can monitor the likelihood of the specific species returning to Brazil due to favourable conditions.
- We can monitor migration patterns of the species. The mosquito is most likely to migrate to areas that are most supportive of their reproductive needs.
- By looking at the climatic conditions we can accesses behavioural patterns of the species and understand evolutionary pathways along with the forces that drive these changes. From what we have leant from evolutionary studies is that most evolution changes are driven by the species need to adapt to aa new environment. The immigration from Africa to Brazil is a major change in regional condition and by looking at the change in the species behaviour we can collarette it to the change in climate observed by climate models.
- We can study what type of climate to malaria diseases thrive in. Most diseases are known to thrive more in specific climatic conditions, by using climate models we can study what areas are more likely to be impacted, ore by the malaria disease.

Cons

- They do not factor in the effect of humans on natural environments. Only natural system is considered as impactors of climate and any significant impact that humans have will not be consider. Industrial emissions do play a role in temperature increases in certain areas of Brazil and this tempura increases are not counted for when looking at the mosquito's ability to survive.
- This models/prediction are centred of theoretical ideologies. When looking back 1 000 years ago we could have not accurately predicted what the climate was going to look like today, because throughout the course of the years the planet has changed drastically compared to what we could have predicted. Just like how we might find species of Anopheles surviving in climate conditions that were deemed unfavourable by climate models.

Are these methods biological sound?

No, From recent studies, we have established that the climate is not only affect buy biotic factors but it is also affected by abiotic factors that most include human activity like deforestation, land scape changes, emission of green house gasses along with some other activities. Climate based models do not take into consideration what effect human have on the climate. One thing I would change about these systems is that I would measure the frequency of human activities that impact climate change and interbreed these frequencies to the system to get more accurate results that include both abiotic and biotic factors.

Results

Figure 1 : climex generated graphs that shows the relative humidity in %, temperature in degrees Celsius and rainfall in mm of Dakar and Natal annually.

The bar graph shows the chance in rainfall of Dakar and Natal throughout the year, we see that the rainfall in Natal slowly increases begging of the year then decreases later in the year compared to that of Dakar in blue that only experiences a spike in August. The slide line graph is the relative humidity at 9 am for most of the year Natal has a higher relative at that time. The dashed line represents the RH at 3pm, again Natal has a higher relative humidity at this time. The bottom-line graph shows the minimum and maximum temperature of both regions throughout the year. The minimum temperature of both regions is above 10 degrees Celsius, and the maximum temperature is below 40 degrees Celsius during the course of the year.

Figure 2: climex generated graphs that shows the relative humidity in %, temperature in degrees Celsius and rainfall in mm of Dakar and Monte Santo annually.

The bar graph shows the chance in rainfall of Dakar and Monte Santol throughout the year, we see that the rainfall of Monte Santo is fairly low with most of the rainfall being under 100mm throughout the year, compared to that of Dakar in blue that only experiences a spike in August. The slide line graph is the relative humidity at 9 am for most of the year Dakar has a higher relative at that time. The dashed line represents the RH at 3pm, again Dakar has a higher relative humidity at this time. The bottom-line graph shows the minimum and maximum temperature of both regions throughout the year. The minimum temperature of both regions is above 10 degrees Celsius, and the maximum temperature is below 40 degrees Celsius during the course of the year.

Figure 3: map of Brazil highlighting the annual rainfall in different regions.
Areas in the west northern regions experience more rainfall with annual temperatures of above 3100mm. this map was put to emphasis the different climatic conditions experienced in Brazil.

Figure 4: graph showing number of malaria cases in Limpopo province arranged by year and moths from 2005 to 2010.
From the jump we notices spikes in malaria infection in months that are related to hot temperatures and high relative humidity such a December to April. In months that are associated with cold temperatures and low relative humidity such as July to September we see small numbers of malaria infection. This graphs gives us a picture of the type of climate conditions we would expect the most malaria infections.

Discussion

Anopheles Gambiae originated from Africa, it travelled to Brazil using mail carrying ships. To truly understand the eradication of this species we need to first understand its habitat conditions and their behavioural patterns. A study was done by Isaac A. Hinne, et al to study diversity of Anopheles mosquito larval habitats in different ecological regions, from this study it was concluded that this species prefers moist environments as the species was abundant in rainy seasons than dry seasons (Isaac A. Hinne, 2021). When looking at the climatic conditions that Brazil experiences we see that that the different regions of the country experience different climate conditions. Some areas experience more rainfall and high temperatures while some areas will experience low rainfall with low temperatures "see figure 3". This ultimately impacts the strategy of the authorities of how much effort would be needed to be put in different region, due to some regions being more prone to more infestation than others. This is observed in the first two figures, Natal rainfall slow increase in the beginning of the year and gradually decreases during the mid-year. While rainfall in Dakar only experiences a spike in August and September "see figure1". This would mean that Natal would be more prone to infestation beginning of the year while Dakar would be more prone to infestation of the mosquito in August and September.

For mosquito to successfully breed they need a blood meal, a pool of water and high temperatures. Depending on the availability of food and the surrounding temperature, the larval stage lasts roughly seven days. To survive the larvae, need water and the larvae will perish if the water source dries up. The conditions in Brazil made it possible of Anopheles Gambiae to reproduce larvae and for the larvae to reach full adult stage. What we observe is that most of the regions of Brazil do not experience significantly high rainfall and temperature through most of the year except for Natal as it has significant rainfall for most of the year "see figure 1 and 2". What this means is that in areas like Dakar and Monte Santo the species would experience months of little to no reproduction due to dry seasons, making it easier for the Brazil authorities to eradicate this species during the months of little to no reproduction. This will slow down the spread of malaria " see figure 4", in this season's medical stuff were not overwhelmed by the number of patients they had to cater for, medical services were more reliable. The cost of treatment for malaria in the dry season was not as heavy on government officials as it probably was in the wet seasons, this allowed for government to focus on other important issues outside from buying Paris green.

When looking at the relative humidity of different areas in Brazil it would have been fairly easy to identify areas that probably would be hotspots for breeding sites, we already know that larvae to grow to reach imago stage they need water. The relative humidity of the different regions would have emphasised areas in which we expect water pools to form. From figure 2 we see that the relative humidity of Dakar throughout the at 3 pm is higher than that of Monte Santo and the

temperature of both regions are below 40 degrees Celsius as its maximum and above 10 degrees Celsius as its minimum “see figure 2”. The main strategies of the eradication of the Anopheles Gambiae in Brazil involved killing the larvae, with data such as the relative humidity and temperature, successfully killing larvae by identifying water pools was an easier approach. Whilst in most countries in Africa they experience a lot of the same climate conditions making it lot harder to identify possible larvae breeding sites.

Paris green is an effective pesticide that was mostly used in the eighteen centuries, a paper was done by Malar J in 2018 where he studied the strategies of malaria control and prevention. When a Paris Green powder was sprayed on a body of water that was infested with anopheline larvae, 24 hours after the pesticide was sprayed approximately 98-100% of the larvae was killed (J.Malar, 2018). This makes Paris Green one of the most effective pesticides of that time and the subsequent use of the pesticide to be beneficial in the battle with Anopheles Gambiae in Brazil. But later studies showed Paris Green as less advantageous, in a paper that L.W Hackette wrote he explained that Paris green in a natural seen is less effective due to disturbance by natural vegetation on the water body, it was toxic to humans and it showed that this pesticide has no effect on eggs and pupae of Anopheles (Hackette, 2023).

Conclusion

In 1930 when the Anopheles Gambia was discovered the first effort of eradication gave, he Brazil government falls hope, when later it was found that 6 years later the species has spread throughout the major regions of Brazil. With their mission to target larvae using pesticides they were able to remain malaria free for 20 years. This victory was a team effort between the government of Brazil and the climate, the climate played a major role in the eradication of Anopheles Gambiae in Brazil in the 1930s, from just climate conditions the government were able to detect how much effort would be needed in each region, what time of the year are the regions less venerable to malaria infections, in dry seasons they were able to financially recover and number of malaria cases decreased, by just assessing the climate conditions we could tell what areas do the mosquito deem as favourable for breeding. Even though the use of Paris Green was effective in larvae it has a lot of disadvantages such its toxicity levels, it is not effective in the egg and pulpal stages and it was shown to be less effective in natural setting with vegetation disturbances.

All the figures referred to in this document will be found on figures file

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